

What is a Solar Cell

A solar cell is the building block of a solar panel. Technically speaking, a solar cell is a semiconductor diode that converts light energy into electrical energy via the photoelectric effect. It is extremely thin – roughly 100 to 180 micrometers thick, which is about the thickness of a single sheet of heavy-duty paper. Meanwhile, it has a comparatively larger surface area, with a common industry standard of 7.2 x 7.2 inches for M10 wafers or 8.3 x 8.3 inches for G12 wafers. This larger surface area enables the solar cell to achieve maximum sunlight absorption.

The formation of Depletion Region in Diode

The most basic structure of a solar cell is the diode consisting of the **p-type material** and the **n-type material**. 99.99% of both materials is made up of pure silicon. In 100% pure silicon, each silicon atom will have 8 electrons on their outermost shell, enabling it to achieve a chemical stability state called octet. However, because the p-type material is doped with 0.01% of a specific atom that has one electron less than silicon, the silicon structure will contain some atoms with one electron less, creating a **hole**. On the other hand, since the n-type material is doped with 0.01% of another specific atom that has one electron more than silicon, the silicon structure will contain some atoms with one extra electron, forcing the electron out from the atomic structure that flows freely in the n type-material, which is called **free electron**. At this point, each material is **still electrically neutral** – each material's number of electrons (the negative charge particle) is equal to each's number of protons (the positive charge particle in the atom's nucleus).

When both materials are attached together, the free electrons in the n-type will diffuse into the p-type; the holes in the p-type will diffuse into the n-type. This causes the n-type to lose its electrons and the p-type to gain extra electrons, forming a **depletion region** (sometimes called p-n junction) at the middle between them where there is an electric field in the direction from the positively charged n-type to the negatively charged p-type (electron is negatively charged). **Figure 1** below illustrates the structure of the depletion region.

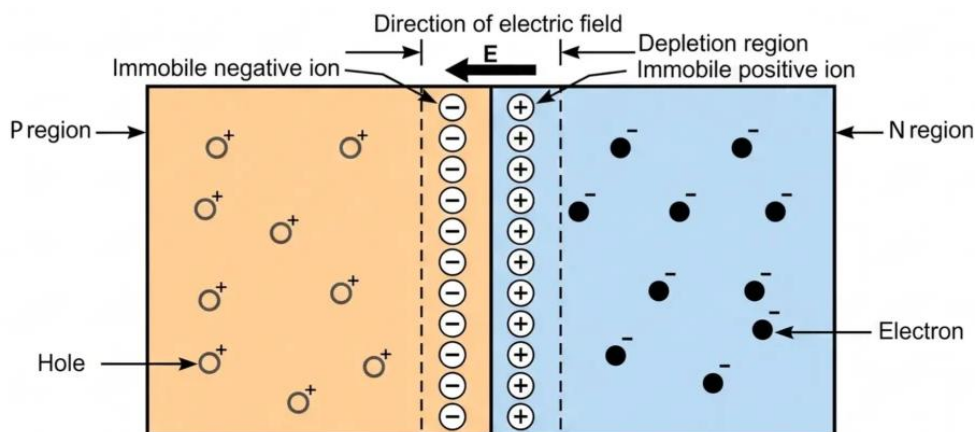


Figure 1: The formation of depletion region

Source: <https://www.hackatronic.com/unbiased-pn-junction-depletion-region-and-barrier-potential/>

Converting Light into Electricity

In the photoelectric effect, a phenomenon in which electromagnetic radiation (light) causes the emission of electrons from a material, light is imagined as a vast number of light particles called **photons**, travelling at the speed of light. Each photon has a specific energy, depending on the light frequency. During the daytime, when photons strike on the solar cell, some will be reflected, some will penetrate, and the remaining will transfer their energy to the electrons. There is one requirement for energy transfer: the photons' energy must be greater than the **material's bandgap threshold**. If the requirement is satisfied, the electrons will be energized and break free off their bonding with the atom, leaving a hole on its original spot and forming an **electron-hole pair**. As electrons are negatively charged, they will be drifted by the force of the electric field, against its direction, toward the end of the n-type material. The positive charged holes will also be carried away by the electric field, towards its direction, reaching the end of p-type material. Throughout this process, there might be a chance of **recombination** of the electrons with the holes, resulting in energy waste. When a conductor is connected to both ends of the solar cell, electron will flow through the conductor to the other end of the solar cell, rejoining the hole and settle off. By putting a light bulb in the middle of the flow, the charge carriers will release their energy to the light bulb, resulting in light emission.

Structure of the Solar Cell

As shown in **Figure 2**, the top layer of the solar cell is an anti-reflective coating, designed to minimize reflection while maximizing absorption. To protect the thin fragile cell from impact and water erosion, a durable and weatherproof tempered glass is covered on top. This is followed by the core structure discussed above: the conductor at the n-side, n-type silicon, depletion region, p-type silicon, and the conductor at the p-side. Due to the higher mobility of electrons, they need to be harvested into the conductor as soon as possible before recombination happens. Therefore, the n-type is placed on top of the p-type to allow the freed electron to move to the exit door (conductor) as soon as possible.

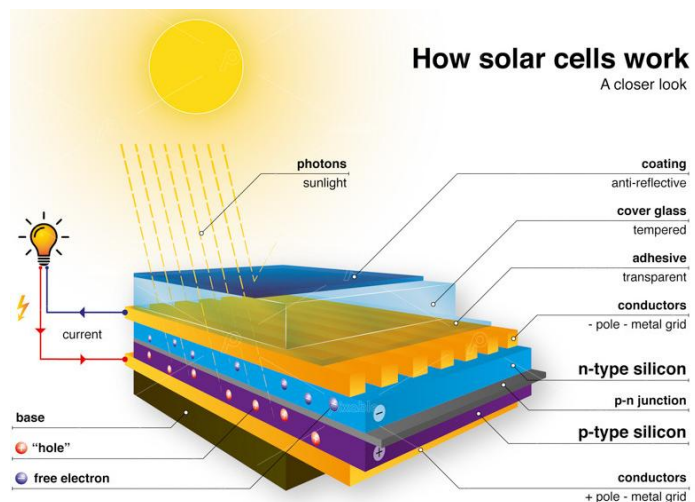


Figure 2: The structure of a solar cell

Source: <https://storage.googleapis.com/dpnuujgqovxline/how-does-a-solar-technology-work.html>

Conclusion

The design of a solar cell is an intricate process that involves semiconductor physics, photoelectric effect, and environmental consideration. As renewable energy has become a defining topic of the 21st century, decades of massive investment have finally made solar panels financially viable for utility and rooftop installation. This progress brings humanity closer to our clean energy goal.